SAT Math

Systems of Linear Equations 1

Question #	ID		Answer
1.1	b86123af		A

Hiro and Sofia purchased shirts and pants from a store. The price of each shirt purchased was the same and the price of each pair of pants purchased was the same. Hiro purchased 4 shirts and 2 pairs of pants for $86, and Sofia purchased 3 shirts and 5 pairs of pants for $166. Which of the following systems of linear equations represents the situation, if *x* represents the price, in dollars, of each shirt and *y* represents the price, in dollars, of each pair of pants?

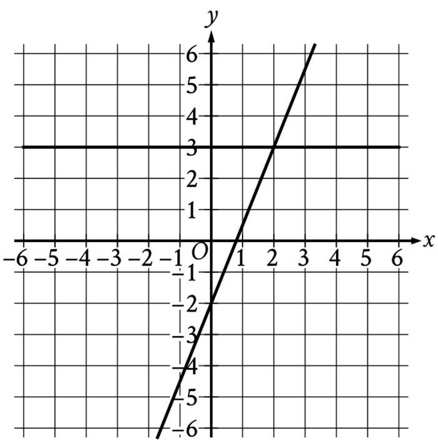
A. $\begin{aligned} 4x+2y&=86 \\ 3x+5y&=166 \end{aligned}$

B. $\begin{aligned} 4x+3y&=86 \\ 2x+5y&=166 \end{aligned}$

C. $\begin{aligned} 4x+2y&=166 \\ 3x+5y&=86 \end{aligned}$

D. $\begin{aligned} 4x+3y&=166 \\ 2x+5y&=86 \end{aligned}$

Question #	ID		Answer
1.2	b0fc3166		C



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

A. $(0, 3)$

B. $(1, 3)$

C. $(2, 3)$

D. $(3, 3)$

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Question #	ID		Answer
1.3	dba8d38a		A

A petting zoo sells two types of tickets. The standard ticket, for admission only, costs $5. The premium ticket, which includes admission and food to give to the animals, costs $12. One Saturday, the petting zoo sold a total of 250 tickets and collected a total of $2,300 from ticket sales. Which of the following systems of equations can be used to find the number of standard tickets, *s*, and premium tickets, *p*, sold on that Saturday?

A. $\begin{aligned} s+p&=250 \\ 5s+12p&=2{,}300 \end{aligned}$

B. $\begin{aligned} s+p&=250 \\ 12s+5p&=2{,}300 \end{aligned}$

C. $\begin{aligned} 5s+12p&=250 \\ s+p&=2{,}300 \end{aligned}$

D. $\begin{aligned} 12s+5p&=250 \\ s+p&=2{,}300 \end{aligned}$

1.4 aff28230 — Answer: **D**

$$x = 10$$
$$y = x + 21$$

The solution to the given system of equations is (x, y). What is the value of y?

A. 2.1

B. 10

C. 21

D. 31

1.5 8abed0fb — Answer: **B**

$y = 2x + 3$
$x = 1$

What is the solution (x, y) to the given system of equations?

A. $(1, 2)$

B. $(1, 5)$

C. $(2, 3)$

D. $(2, 7)$

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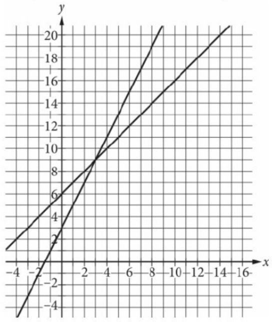
Question #	ID		Answer

1.6 e1259a5a

A system of two linear equations is graphed in the xy-plane below.



Which of the following points is the solution to the system of equations?

A. $(3,9)$

B. $(6,15)$

C. $(8,10)$

D. $(12,18)$

Answer: **A**

1.7 ca9bb527

$$y = 4x - 9$$
$$y = 19$$

What is the solution (x, y) to the given system of equations?

A. $(4, 19)$

B. $(7, 19)$

C. $(19, 4)$

D. $(19, 7)$

Answer: **B**

1.8 ece00725

Connor has c dollars and Maria has m dollars. Connor has 4 times as many dollars as Maria, and together they have a total of $\$25.00$. Which system of equations represents this situation?

A. $c = 4m$
$c + m = 25$

B. $m = 4c$
$c + m = 25$

C. $c = 25m$
$c + m = 4$

D. $m = 25c$
$c + m = 4$

Answer: **A**

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Question #	ID		Answer

1.9 ee031767

A dance teacher ordered outfits for students for a dance recital. Outfits for boys cost $26, and outfits for girls cost $35. The dance teacher ordered a total of 28 outfits and spent $881. If *b* represents the number of outfits the dance teacher ordered for boys and *g* represents the number of outfits the dance teacher ordered for girls, which of the following systems of equations can be solved to find *b* and *g* ?

A. $26b + 35g = 28$
$b + g = 881$

B. $26b + 35g = 881$
$b + g = 28$

C. $26g + 35b = 28$
$b + g = 881$

D. $26g + 35b = 881$
$b + g = 28$

Answer: **B**

1.10 cd33b015

$$x + y = 20$$
$$2(x + y) + 3y = 85$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

A. 10

B. 15

C. 60

D. 65

Answer: **B**

1.11 0d1dca87

$$3x + y = 29$$
$$x = 2$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

Answer: **23**

Question #	ID		Answer
1.12	0df106df	An online bookstore sells novels and magazines. Each novel sells for \$4, and each magazine sells for \$1. If Sadie purchased a total of 11 novels and magazines that have a combined selling price of \$20, how many novels did she purchase? A. 2 B. 3 C. 4 D. 5	B
1.13	7d89376f	A discount airline sells a certain number of tickets, x, for a flight for \$90 each. It sells the number of remaining tickets, y, for \$250 each. For a particular flight, the airline sold 120 tickets and collected a total of \$27,600 from the sale of those tickets. Which system of equations represents this relationship between x and y ? A. $\begin{cases} x+y=120 \\ 90x+250y=27{,}600 \end{cases}$ B. $\begin{cases} x+y=120 \\ 90x+250y=120(27{,}600) \end{cases}$ C. $\begin{cases} x+y=27{,}600 \\ 90x+250y=120(27{,}600) \end{cases}$ D. $\begin{cases} 90x=250y \\ 120x+120y=27{,}600 \end{cases}$	A
1.14	17f176ec	A movie theater charges \$11 for each full-price ticket and \$8.25 for each reduced-price ticket. For one movie showing, the theater sold a total of 214 full-price and reduced-price tickets for \$2,145. Which of the following systems of equations could be used to determine the number of full-price tickets, f, and the number of reduced-price tickets, r, sold? A. $f+r=2{,}145$ $11f+8.25r=214$ B. $f+r=214$ $11f+8.25r=2{,}145$ C. $f+r=214$ $8.25f+11r=2{,}145$ D. $f+r=2{,}145$ $8.25f+11r=214$	B

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Question #	ID		Answer
1.15	44d65912	Angela is playing a video game. In this game, players can score points only by collecting coins and stars. Each coin is worth c points, and each star is worth s points.	B

- The first time she played, Angela scored 700 points. She collected 20 coins and 10 stars.
- The second time she played, Angela scored 850 points. She collected 25 coins and 12 stars.

Which system of equations can be used to correctly determine the values of c and s ?

A. $$\begin{aligned} 10c + 20s &= 700 \\ 12c + 25s &= 850 \end{aligned}$$

B. $$\begin{aligned} 20c + 10s &= 700 \\ 25c + 12s &= 850 \end{aligned}$$

C. $$\begin{aligned} 20c + 700s &= 10 \\ 25c + 850s &= 12 \end{aligned}$$

D. $$\begin{aligned} 700c + 20s &= 10 \\ 850c + 25s &= 12 \end{aligned}$$

1.16 4b76c7f1 — Answer: **C**

$$\begin{aligned} 2x + 7y &= 9 \\ 8x + 28y &= a \end{aligned}$$

In the given system of equations, a is a constant. If the system has infinitely many solutions, what is the value of a ?

A. 4

B. 9

C. 36

D. 54

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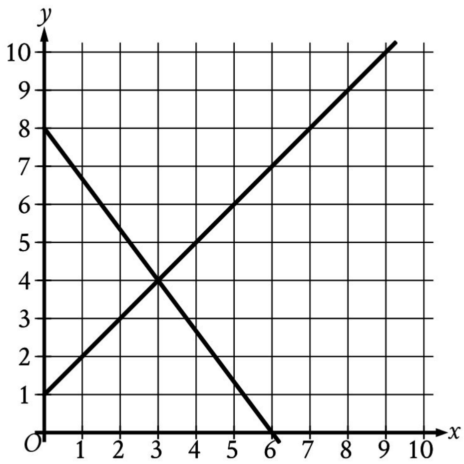
Question # **ID** **Answer**

1.17 e6545fa8 **B**



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

A. $(2, 3)$

B. $(3, 4)$

C. $(4, 5)$

D. $(5, 6)$

1.18 f5563c26 **A**

$$y = 4$$
$$x = y + 6$$

The solution to the given system of equations is (x, y). What is the value of x?

A. 10

B. 6

C. 4

D. 2

1.19 608eeb6e **C**

$$5x = 15$$
$$-4x + y = -2$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

A. -17

B. -13

C. 13

D. 17

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Question #	ID		Answer
1.20	317e80f9	$$x + y = 18$$ $$5y = x$$ What is the solution (x, y) to the given system of equations? A. $(15, 3)$ B. $(16, 2)$ C. $(17, 1)$ D. $(18, 0)$	A
1.21	4fb8adf7	$$4x - 3y = 5$$ $$x = 8$$ What is the solution (x, y) to the given system of equations? A. $(8, 9)$ B. $(8, -24)$ C. $(8, -9)$ D. $(8, 24)$	A